$R(D_s)$ signal MC: kinematic-fit schemes, B_s hypotheses and what the BDT gains from each input

Summary of `compare_fits` `cmp_v1`–`cmp_v3` and the full-statistics `hypinfo` study

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The answers in one page

Every statement about an input is only meaningful together with *what else is already in the fit*. The table gives each conclusion with that condition. The sections after it give the evidence.

question	answer	condition, evidence
Which vertex-fit scheme?	**`free`** (no mass constraint). Decided.	All four schemes give the same physics within 1%; the ϕ constraint loses 32–35% of the signal, the D_s constraint 5% (Sec. 2).
Which B_s momentum hypothesis?	**`xyz_bs`** as the primary one.	Best alone in every test; `coll` is a single variable in disguise (Sec. 3).
Does `coll` add anything?	**No**, in any configuration.	$\leq 1\%$ on top of `xyz_bs`, with or without other inputs (Sec. 4).
Does `math` add anything?	**Depends on the other inputs.**	BDT with kinematics only: $-7.5 \pm 0.8\%$ on $\sigma R(D_s)$. BDT with context: $-0.7 \pm 0.3\%$, i.e. nothing. Binned 2D fit: -11%, not yet robust (Sec. 4, 5).
Is that gain specific to `math`?	**No.**	The raw track kinematics next to `xyz_bs` give the same gain (-6.5%).
Do the photon hypotheses (q^2, m^2_{miss} with $D_s\gamma$) add anything?	**No**, once the photon *quality* variables are in.	$-0.4 \pm 0.3\%$ on top of context. The photon quality variables themselves are the largest single gain for R (-35% in `cmp_v2`).
What does matter?	The **context** variables (topology, photon quality, kinematics, isolation).	-45% on $\sigma R(D_s)$, -38% on $\sigma R(D_s^*)$, -15% on $\sigma_{\tau/\mu}$ vs `xyz_bs` alone.
Can the method see a real gain?	**Yes.**	Two positive controls give -27% on $\sigma R(D_s)$.

1 How the comparisons are made

Samples. UL18 $B_s \to D_s^{(*)}\ell\nu$ signal MC (first 500 files), selection: truth-matched signal of the four channels, correct particle species, trigger path and muon match, one candidate per event.

study	input	$D_s\mu\nu$	$D_s^*\mu\nu$	$D_s\tau\nu$	$D_s^*\tau\nu$
`cmp_v1`, `cmp_v2`	`rds_prod.root`, 17% read	114 684	347 552	5 986	15 680
`cmp_v3`	dsc production, 27% of 239 chunks	114 531	346 149	5 780	15 543
`hypinfo_full`	dsc production, all 500 chunks	929 478	2 822 510	48 024	126 751

Figure of merit. Every number is the expected relative uncertainty from a *binned Poisson fit of the four signal yields with perfect templates*: no background, no systematics, no template statistics. The τ channels are weighted to $R(D_s) = 0.299$, $R(D_s^*) = 0.244$ and the sample is scaled to 10^5 $D_s\mu\nu$. Three quantities:

- $\sigma_{\tau/\mu}$: two yields (D_s and D_s^* merged), so it measures τ/μ separation alone;

- $\sigma R(D_s)$, $\sigma R(D_s^*)$: four yields, so they also need D_s told apart from D_s^*. This is the hard part: $\sigma R(D_s)$ is 15–28 times its counting limit, $\sigma_{\tau/\mu}$ only about 6 times.

These numbers *rank* inputs; they are not the sensitivity of the analysis.

What is binned. For observables (Sec. 3, 5): equal-population bins, 12 per axis in 2D, 6 in 3D. For a BDT (Sec. 4): the four-class output is reduced to the plane $p_\tau = P(D_s\tau) + P(D_s^*\tau)$, $p_* = P(D_s^*\mu) + P(D_s^*\tau)$, binned in 10×10 ("b10", `cmp_v2`) or 20×20 ("b20", `hypinfo`) equal-population bins. b10 loses about 10% against b20. An unbinned version was tried and found unstable; it is not used.

How precise the figures are. Between two runs on different events, $\sigma R(D_s)$ moves by 5–12% (`cmp_v2` vs `cmp_v3`). Between two input sets on the *same* events, a paired bootstrap gives uncertainties of 0.1–1%. So: differences within one study are meaningful at the per-cent level; absolute values and differences between studies are not, below $\approx 10\%$.

2 Fit schemes

Four kinematic-fit schemes are run for every candidate. They differ only in the mass constraints:

scheme	$m(KK) = m_\phi$	$m(KK\pi) = m_{D_s}$	note
`cons`	yes	yes	
`phic`	yes	no	the AN choice
`dsc`	no	yes	added for `cmp_v3`
`free`	no	no	**chosen**

Table 1: Fit quality on truth-matched signal (μ channels), `cmp_v3`. Vertex cuts: $P(V_\phi) > 1\%$ and $P(V_{D_s}) > 1\%$. A calibrated fit has $\chi^2/\text{ndf} \approx 1$ and 1% of candidates below $P = 1\%$.

scheme	V_ϕ χ^2/ndf	V_ϕ $P < 1\%$	V_{D_s} χ^2/ndf	V_{D_s} $P < 1\%$	vertex-cut efficiency	$p_T(D_s)$ σ_{68}
`cons`	9.98	0.319	3.33	0.100	0.645	0.56%
`phic`	9.98	0.319	1.06	0.006	0.679	1.02%
`dsc`	0.94	0.005	2.45	0.046	0.950	0.46%
`free`	0.94	0.005	0.93	0.001	**0.995**	0.80%

- **The ϕ constraint is wrong by construction.** The K^+K^- mass spread is dominated by the ϕ natural width ($\Gamma_\phi \approx 4.2\,\text{MeV}$), which a constraint built on track errors ignores. Its χ^2 term averages ≈ 19 per degree of freedom. With the $P(V_\phi)$ cut it removes a third of the signal. It also pulls the kaons: $m(D_s)$ resolution 13.1 vs 10.6 MeV and $p_T(D_s)$ 1.02% vs 0.80% in `phic` vs `free`.
- **The D_s constraint is mildly miscalibrated:** its term averages ≈ 4 instead of 1, and $P(V_{D_s}) > 1\%$ removes 5% of the signal. The mass-uncertainty demonstrator (separate note, 778k truth-matched candidates) explains it: $(m_{KK\pi} - m_{D_s})/\sigma_m$ from the track covariances has $\langle \text{pull}^2 \rangle = 4.3$, of which the Gaussian core contributes only 1.15^2; the rest is non-Gaussian tails (3.4% beyond 3σ, 1% beyond 5σ), partly FSR. Rescaling the momentum errors would not fix it. (The "57% fail" quoted from the smoke test in `cmp_v2` was over all candidates, combinatorial ones included; it does not describe signal.)
- **Both losses are equal in the four channels**, so they do not bias R; they only cost statistics ($\sqrt{0.995/0.65} \approx 1.24$ in precision for the ϕ constraint).
- **The physics is the same in every scheme.** The B_s momentum is uncertain at 10–20% (direction from the vertices, magnitude from the visible momentum); a better D_s momentum ($< 1\%$ in every scheme) does not show:

Table 2: `xyz_bs` observables per scheme, `cmp_v3`. 3D: binned $(q^2, m^2_{\text{miss}}, E^*_\mu)$. BDT: `xyz_bs` observables + topology + photon + common, 4 folds, b10.

	σ_{68} resolution		3D binned fit			BDT		
scheme	q^2 [GeV^2]	m^2_{miss} [GeV^2]	$\sigma_{\tau/\mu}$	$\sigma R(D_s)$	$\sigma R(D_s^*)$	$\sigma_{\tau/\mu}$	$\sigma R(D_s)$	$\sigma R(D_s^*)$
`cons`	1.753	0.917	0.0861	0.699	0.265	0.0775	0.436	0.197
`phic`	1.765	0.921	0.0863	0.698	0.264	0.0777	0.438	0.196
`dsc`	1.744	0.910	0.0860	0.693	0.263	0.0772	0.436	0.196
`free`	1.758	0.913	0.0862	0.697	0.264	0.0773	0.431	0.194

Scheme: `free`. Same physics as every other scheme within 1%, and 99.5% of the signal through the vertex cuts against 95% (`dsc`) and 65–68% (`cons`, `phic`). A BDT fed with the inputs of all four schemes together (215 features) gains nothing either.

3 B_s momentum hypotheses

The B_s momentum is not measured; four hypotheses build it from the visible $D_s\mu$ and the PV$\to V_{B_s}$ direction, and every observable (q^2, m^2_{miss}, E^*_μ, angles, ...) is computed in each.

`xyz_bs`	direction PV$\to V_{B_s}$, magnitude $\lvert p_{B_s}\rvert = \lvert p_{D_s\mu}\rvert\, m_{B_s}/m_{D_s\mu}$
`coll`	along the visible $D_s\mu$ momentum, same magnitude
`math1`, `math2`	the two analytic solutions for one massless neutrino and the PV$\to V_{B_s}$ direction; $m^2_{\text{miss}} = 0$ by construction; real for 68–88% of candidates
`math_oracle`	the `math` solution closer to the truth (a positive control, not usable in data)

Table 3: Per hypothesis, `free` scheme, 3D binned fit ($q^2, m^2_{\text{miss}}, E^*_\mu$), `cmp_v1`.

hypothesis	q^2 σ_{68} [GeV^2]	q^2 bias	E^*_μ σ_{68}	$\sigma_{\tau/\mu}$	$\sigma R(D_s)$	$\sigma R(D_s^*)$
`xyz_bs`	1.75	−0.02	0.164	**0.085**	**0.73**	**0.30**
`math1`	2.20	+0.84	0.173	0.102	0.81	0.35
`math_avg`	2.11	+0.42	0.169	0.102	0.81	0.37
`math2`	2.56	−0.07	0.201	0.111	1.24	0.57
`coll`	2.22	+1.12	0.228	0.112	7.6	2.9
`math_oracle`	1.51	+0.28	0.111	0.104	0.65	0.29
generator level	–	–	–	0.045	0.24	0.10

- **`coll` is one variable.** With $p_{B_s} \propto p_{D_s\mu}$, every invariant depends on $m(D_s\mu)$ only: its q^2, m^2_{miss}, E^*_μ are 100% rank-correlated, and a 2D/3D fit cannot separate D_s from D_s^*.
- **`math` loses m^2_{miss}**, the strongest single τ/μ variable (m^2_{miss} of `xyz_bs` alone gives $\sigma_{\tau/\mu} = 0.081$). It reconstructs other quantities better than `xyz_bs`: $p_T(B_s)$, p_T^{miss}, the angles. That is why it was worth testing as an *additional* input (Sec. 4).
- The reconstruction loses a factor ≈ 1.9 in $\sigma_{\tau/\mu}$ and ≈ 3 in $\sigma R(D_s)$ relative to generator level.

4 BDT performance versus inputs

Set-up. One four-class gradient-boosted classifier (scikit-learn `HistGradientBoosting`, balanced classes, out-of-fold predictions from 4 folds) per input set, `free` scheme. Two campaigns:

	`cmp_v2` (ablation)	`hypinfo_full`
events	120k per μ channel, all τ ($\approx$ 6k $D_s\tau$)	everything (48k $D_s\tau$)
training	stopped after ≈ 45 iterations	to convergence (200–2000 iterations)
binning of the output plane	10×10	20×20
uncertainty	none (scatter $\approx 1\%$)	paired bootstrap, 50 replicas

The input groups. Each hypothesis contributes the same 9 observables computed in its B_s rest frame: q^2, m^2_{miss}, p_T^{miss}, E^*_μ, $E^{\#}_\mu$, $\cos\theta(\phi, D_s)$, $\cos\theta(\mu, W)$, χ, $p_T(B_s)$.

group	n	content
`xyz`	9	the 9 observables of `xyz_bs` (`q3`: only q^2, m^2_{miss}, E^*_μ)
`coll`	9	the same for `coll`
`math`	18	the same for `math1` and `math2`
`raw`	13	p_T, η, $\Delta\phi$ to the flight direction of K_1, K_2, π, μ; η of the flight direction
`inv`	9	$\lvert p_{D_s}\rvert$, $\lvert p_\mu\rvert$, $m(KK\pi)$, $m(KK)$, $\cos(D_s,\mu)$, $\cos(D_s,\mathrm{flight})$, $\cos(\mu,\mathrm{flight})$, $(\vec{p}_{D_s}\times\vec{p}_\mu)\cdot\hat{d}$, $\cos\theta(K,\pi)$
`ctx` (context)	29	*topology* (11): μ–D_s DOCA and significance, μ POCA along D_s, μ 3D IP significance, μ distance to the B_s direction, $P(V_{B_s})$, B_s and D_s flight significances, m_{corr}, $p_\perp$, $\cos\theta(K,\pi)$; *photon quality* (3): $p_T(\gamma)$, Δm, E_{coll}/p_T; *common* (15): masses, ΔR's, track p_T's, μ η, μ IP significances, isolations
`γhyp`	4	q^2 and m^2_{miss} with $D_s\gamma$ in place of D_s, for `xyz_bs` and `coll`

Positive controls. To show that the test can detect information when it is there, two sets add an input that is known to carry some: the truth-picked `math` solution (`oracle`), and the generator q^2 smeared by $1\,\mathrm{GeV}^2$.

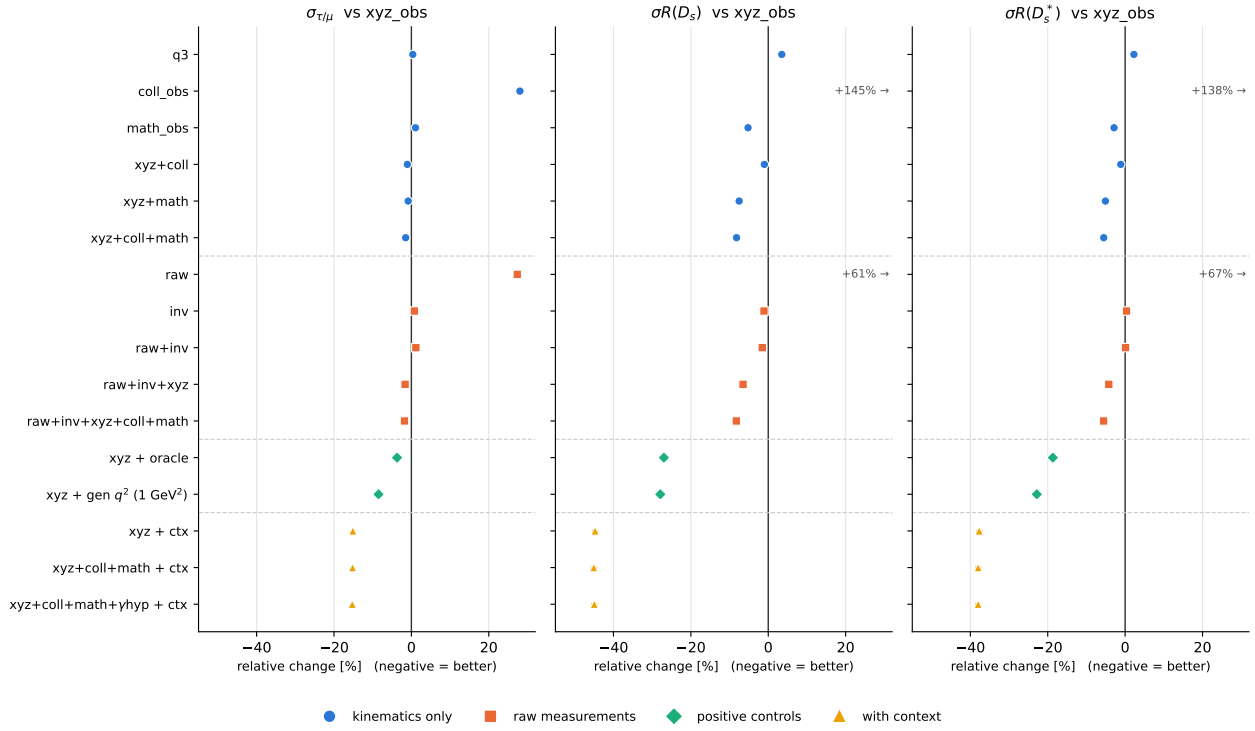


Figure 1: Change of the expected precision relative to the BDT on the 9 `xyz_bs` observables, on the same events (`hypinfo_full`, b20). Error bars: paired bootstrap, mostly smaller than the markers. Off-scale values are printed at the right edge.

Table 4: All `hypinfo_full` BDTs: absolute precision (b20), out-of-fold log-loss (lower is better), and the change relative to `xyz_obs` in per cent.

inputs	n	expected precision (20×20 bins) $\sigma_{\tau/\mu}$	$\sigma R(D_s)$	$\sigma R(D_s^*)$	log-loss	change vs `xyz_obs` (paired bootstrap) τ/μ	$R(D_s)$	$R(D_s^*)$
reference								
xyz_obs	9	0.0771	0.579	0.245	1.0944	–	–	–
kinematics only								
q3	3	0.0774	0.599	0.251	1.0969	$+0.4 \pm 0.1$	$+3.5 \pm 0.8$	$+2.3 \pm 0.7$
coll_obs	9	0.0988	1.419	0.583	1.1912	$+28.0 \pm 0.3$	$+145.1 \pm 9.9$	$+137.7 \pm 8.8$
math_obs	18	0.0779	0.549	0.238	1.0958	$+1.1 \pm 0.1$	-5.2 ± 1.0	-2.9 ± 0.9
xyz+coll	18	0.0763	0.573	0.243	1.0943	-1.0 ± 0.0	-1.0 ± 0.5	-1.1 ± 0.4
xyz+math	27	0.0765	0.535	0.233	1.0889	-0.8 ± 0.1	-7.5 ± 0.8	-5.1 ± 0.7
xyz+coll+math	36	0.0760	0.531	0.232	1.0872	-1.5 ± 0.1	-8.2 ± 0.8	-5.5 ± 0.7
raw measurements								
raw	13	0.0982	0.930	0.409	1.2082	$+27.4 \pm 0.3$	$+60.7 \pm 3.3$	$+66.6 \pm 3.1$
inv	9	0.0777	0.572	0.246	1.1078	$+0.8 \pm 0.1$	-1.1 ± 1.0	$+0.4 \pm 0.9$
raw+inv	22	0.0780	0.570	0.246	1.1102	$+1.1 \pm 0.1$	-1.5 ± 0.9	$+0.1 \pm 0.8$
raw+inv+xyz	31	0.0759	0.541	0.235	1.0874	-1.6 ± 0.1	-6.5 ± 0.8	-4.2 ± 0.7
raw+inv+xyz+coll+math	58	0.0758	0.531	0.232	1.0857	-1.8 ± 0.1	-8.3 ± 0.8	-5.6 ± 0.7
positive controls								
xyz + oracle	18	0.0743	0.423	0.199	1.0683	-3.7 ± 0.1	-27.0 ± 0.9	-18.7 ± 0.8
xyz + gen. q^2 smeared	10	0.0706	0.417	0.189	1.0578	-8.5 ± 0.1	-27.9 ± 0.7	-22.8 ± 0.7
with context								
xyz + ctx	38	0.0655	0.320	0.153	1.0308	-15.1 ± 0.1	-44.7 ± 0.8	-37.7 ± 0.8
xyz+coll+math + ctx	65	0.0654	0.318	0.152	1.0338	-15.2 ± 0.1	-45.1 ± 0.8	-38.0 ± 0.8
xyz+coll+math+γ-hyp + ctx	69	0.0653	0.319	0.152	1.0306	-15.3 ± 0.1	-45.0 ± 0.8	-38.0 ± 0.8

Reading the results in four steps

1. **Kinematics only, no context** (blue in Fig. 1). `coll` adds 1%. `math` adds $7.5 \pm 0.8\%$ on $\sigma R(D_s)$ and $5.1 \pm 0.7\%$ on $\sigma R(D_s^*)$, nothing on τ/μ; its log-loss drop (1.0944 → 1.0889) confirms it. *Here, and only here, `math` helps.*

2. **Is it `math` itself?** (orange). The raw track kinematics next to `xyz` give almost the same gain (-6.5%) with no `math`; adding `math` on top brings it to -8.3%. Alone, the raw kinematics are much worse ($+61\%$): the BDT cannot build B_s-frame quantities itself, but uses raw kinematics as corrections once `xyz` is there. So the information exists in the measurements, `xyz`'s 9 observables leave some of it out, and `math` is one way to recover it.

3. **The method works** (green). The two controls gain 27–28% on $\sigma R(D_s)$ with large log-loss drops. A null result elsewhere therefore means no information, not a blind method.

4. **The realistic BDT** (yellow). With the context variables in, the picture changes:

Table 5: On top of `xyz` + context (38 features), paired bootstrap, b20.

added to xyz + ctx	n	τ/μ	$R(D_s)$	$R(D_s^*)$
coll + math	65	$-0.1 \pm 0.1\%$	$-0.7 \pm 0.3\%$	$-0.4 \pm 0.3\%$
coll + math + photon hypotheses	69	$-0.2 \pm 0.1\%$	$-0.4 \pm 0.3\%$	$-0.4 \pm 0.3\%$

BDT inputs: `xyz_bs` + context. `coll`, `math` and the photon hypotheses add $< 1\%$ once the context is in: what `math` recovers in step 1 is already covered by the context variables, which contain the same kinematics (masses, track p_T's, ΔR's). The context is the large lever: -45% on $\sigma R(D_s)$, -38% on $\sigma R(D_s^*)$, -15% on $\sigma_{\tau/\mu}$.

Consistency with `cmp_v2`. The ablation on 1/8 of the statistics, short training and b10 reached the same conclusions (`math` and `coll` change nothing when removed from the full set) and isolated the groups inside the context: the photon quality variables give the largest gain for R ($\sigma R(D_s)$ 0.571 → 0.384 when added back to everything, -33%), the topology the gain for τ/μ (0.0833 → 0.0759, -9%), of which two thirds is the μ–D_s DOCA significance. $P(V_{B_s})$ is a monotonic function of that DOCA significance (V_{B_s} is a $D_s + \mu$ fit with one degree of freedom): use one of the two, not both. Absolute values differ between the campaigns (full set: $\sigma R(D_s)$ 0.384 in `cmp_v2` b10, 0.356 b10 and 0.319 b20 here), as expected from Sec. 1.

5 Binned fits on observables

The binned fits do not use the context, so they behave like step 1 above: `math` helps.

Table 6: Best 2D (12 bins per axis) and 3D (6 per axis) combinations among the 15 observables of `xyz_bs`, `coll`, `math1`, `math2`, `math_avg`; first row of each block: the best `xyz_bs`-only set.

dim	observables	$\sigma_{\tau/\mu}$	$\sigma R(D_s)$	$\sigma R(D_s^*)$	min MC/bin
2D	xyz:q^2, xyz:m^2_{miss}	0.0841	0.700	0.268	2
2D	xyz:q^2, math_avg:q^2	0.0745	0.625	0.256	3
2D	xyz:m^2_{miss}, math_avg:q^2	0.0857	0.633	0.276	1
3D	xyz:q^2, xyz:m^2_{miss}, xyz:E^*_μ	0.0847	0.731	0.284	1
3D	xyz:q^2, math1:q^2, math2:E^*_μ	0.0809	0.592	0.257	3
3D	xyz:q^2, xyz:E^*_μ, math1:q^2	0.0841	0.618	0.264	2

- q^2 of `xyz_bs` against q^2 of `math_avg` beats (q^2, m^2_{miss}) of `xyz_bs` by 11% on $\sigma R(D_s)$ and on $\sigma_{\tau/\mu}$.
- **Not yet robust:** these configurations have bins with 1–3 MC events, which makes perfect-template precision look better than it is (`cmp_v2` found the same with the photon 4D set), and there is no bootstrap. To check before use: coarser bins or a minimum MC count per bin, and a bootstrap.
- The photon hypotheses gain at most a few per cent in 2D/3D (`cmp_v2`): the photon information is mostly in the quality variables, which a binned fit on q^2-like observables cannot use.

6 Decisions and open items

1. **Scheme `free`**; the ϕ constraint and its $P(V_\phi)$ cut go. All four schemes stay in the ntuple for now.
2. **BDT inputs:** `xyz_bs` observables + context; `coll`, `math`, photon hypotheses not needed. One of $P(V_{B_s})$ or the μ–D_s DOCA significance.
3. **Open:** if the final fit is binned in observables rather than in a BDT output, re-test `xyz_bs`:q^2 × `math_avg`:q^2 with adequate MC per bin.
4. **Understood:** the D_s-constraint χ^2 excess (≈ 4 per dof) comes from mass-resolution tails, not from a global underestimate of the momentum errors (core 13–15% too narrow in MC); see the mass-uncertainty note.

Check yourself

1. A colleague reads "`math` gives -7.5% on $\sigma R(D_s)$" and proposes adding it to the analysis BDT. What do you answer, and which table settles it?

2. Why does `raw` alone do much worse than `xyz_obs`, while `raw+inv+xyz` does better than `xyz_obs`?

3. `cmp_v2` gave $\sigma R(D_s) = 0.384$ for the full set, `hypinfo` 0.319. Is the BDT better now? Name the three differences that each move this number.

4. Why does removing $P(V_{B_s})$ from the full set change nothing, although $P(V_{B_s})$ alone is the strongest τ/μ addition to `xyz_obs`?

5. The ϕ-constraint loss is equal in all four channels. Why does it still matter, and by how much in precision?